

Review for Quiz 1



CAS AR280 O2, Summer Session II, 2026
Tuesday (7/7/2026)

Format for Quiz 1

- Will go live tomorrow, Wednesday, July 8
- Closes **Sunday, July 12, 11:59 PM ET (Boston time)**
- Can only open the quiz ONCE
- Time limit: 45 minutes
- 15 questions (randomized)
- 1 short essay question (randomized)
- 25 points total, each Quiz worth 5% of your total grade
- Will cover all of Modules 1 and 2

Types of Questions

- Multiple Choice
 - Selecting one or multiple correct answers, “most correct” answer
 - Matching
 - Ordering
- Fill in the blank
- All worth 1 pt each, for a total of 15 points

Short Essay

- Format: “You are an archaeologist studying _____ in _____ region.”
 - Randomized questions will give you a specific scenario
 - How would you excavate the site? Why?
 - What methods would you use? Why?
 - What analyses would you perform? Why?
 - How would you interpret the data you get? What is your reasoning? What conclusions can you make?
- Need to answer in complete sentences, no bullet points
- Need to fully explain the logic behind your answers
- Make sure to answer ALL components of the question
- Worth 10 points

Module Study Guides

- How to access:
 - Home Page →
 - Module 1/Module 2 →
 - Will automatically open to the Study Guide: **provides in-depth explanations of terms + examples**
 - In the Table of Contents for the Study Guide, last section will be **“Key Terms”**
 - Provides definitions of each, so you can test yourself

Module 1

-  Module 1 Overview
-  An Introduction to Archaeology and Anthropology
-  The Anthropology of Food and Culture
-  Archaeological Approaches to the Study of Food
-  Methods: Archaeological Survey and Excavation
-  Key Terms
-  Module 1 Wrap Up
-  Food Journal Assignment 1

Week 1: Archaeological and Anthropological Approaches to Food and Culture

Learning Objectives:

- Distinguish the fields of archaeology and anthropology and describe their historical and ongoing relationships
- Define the 4 subfields of anthropology
- Associate key anthropological theories of food with their authors and proponents
- Recognize key questions in the archaeological study of food and drink
- Describe basic methods of archaeological survey and excavation

Week 1: Key Terms

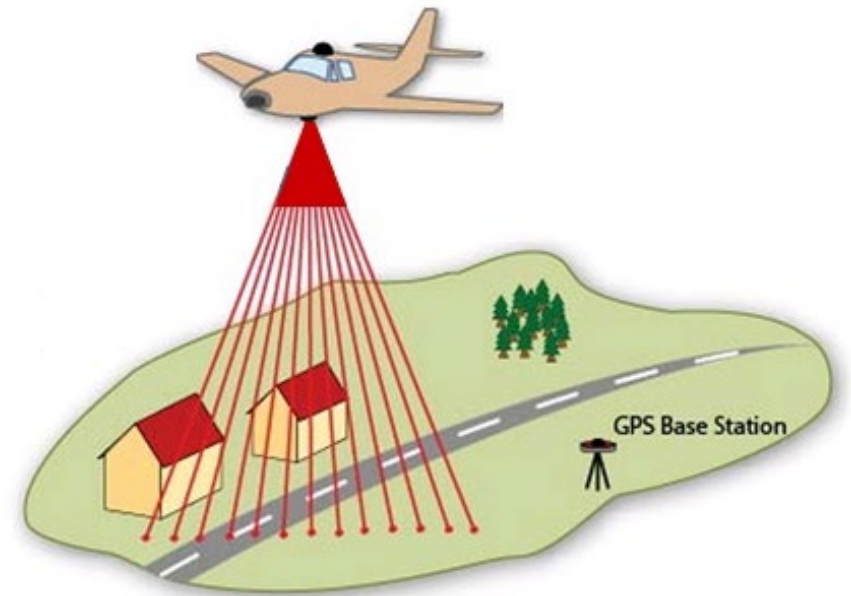
- Archaeology
- Archaeological evidence
- Artifacts
- Chemical residue analysis
- Culture
- Ethnography
- Excavation
- Features
- Flotation
- Geophysics
- Grinding stones
- Historical archaeology
- Inference
- Isotopes
- Lidar
- Material remains
- Materialism
- Microbotanical remains
- Paleoethnobotany
- Photogrammetry
- Phytoliths
- Proxy
- Radiocarbon dating
- Semiotics
- Sites
- Starch granules
- Strata
- Stratigraphy
- Structuralism
- Superposition
- Survey
- Uniformitarianism
- Zooarchaeology

Excavation Methods

- Excavation: the practice of digging carefully and systematically to uncover features, artifacts, and strata/stratigraphy.
- Beginning:
 - **Survey** (*pedestrian or intensive*) for arch. remains/surface collection
 - **LiDAR** (light detection and ranging): precise mapping of ground surfaces using a drone that uses wavelengths of light that bounce off the ground, penetrating vegetation
 - **Geophysics**: set of techniques that often use magnetic fields to identify features such as walls or pits without excavation

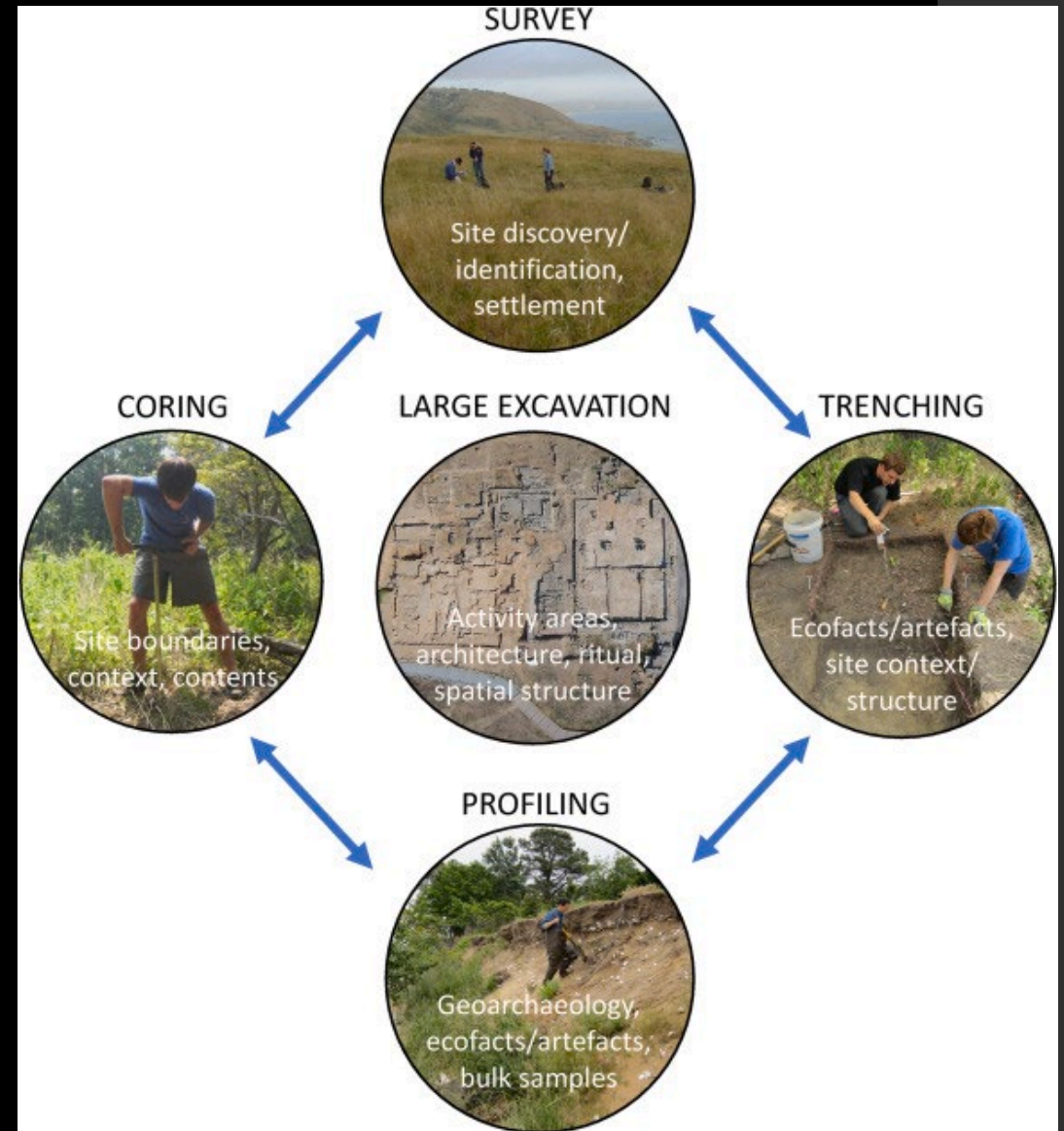


Airborne LiDAR – 3D mapping



Excavation Methods

- **Picking your Research Questions**
(this will affect the samples you take, where you excavate, how deep you dig, etc.)
 - What are the political and social roles of the inhabitants?
 - How was the city laid out?
 - What did they eat, and where was that food grown?
 - How did they produce or trade for the goods they needed?
 - How long has the site been occupied?



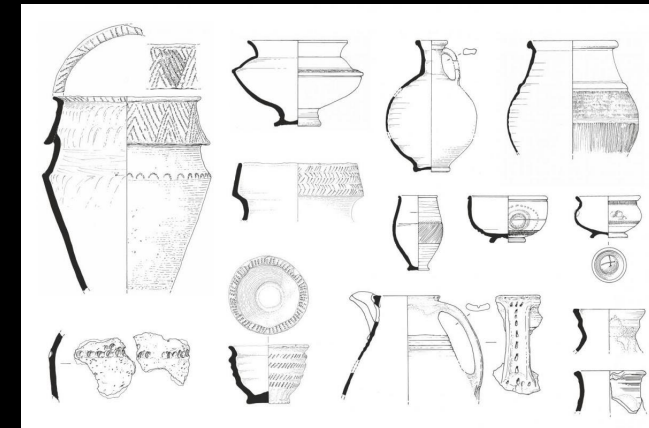
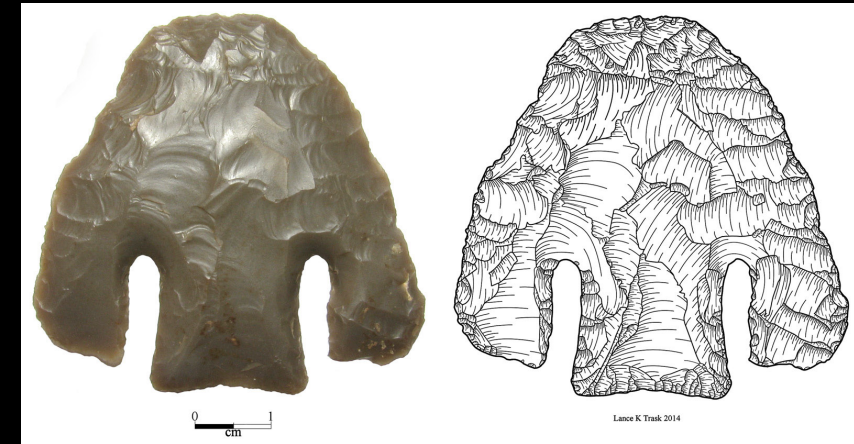
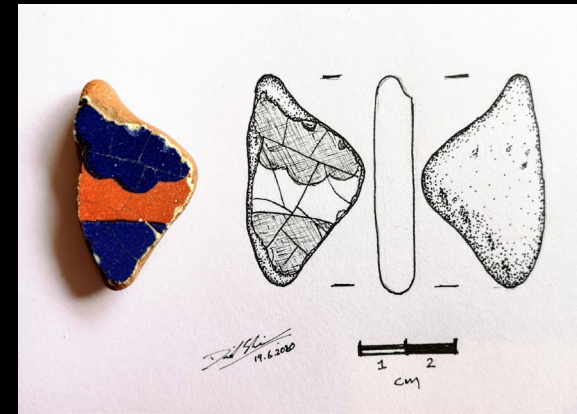
Excavation Methods

- Excavation:
 - Recording and identifying the **stratigraphy** of the site
 - Identify special **features** (hearths, stone tool production areas, food storage areas, trash pits, etc.)
 - Can record these with **pictures and GPS data** to digitally re-create the site in 3D
 - Take any **samples** you need: soil for botanical analysis; screen soil for animal bones, ceramics, and stone tools; blocks of soil for geochemical analysis; carbonized plants for radiocarbon dating, etc.



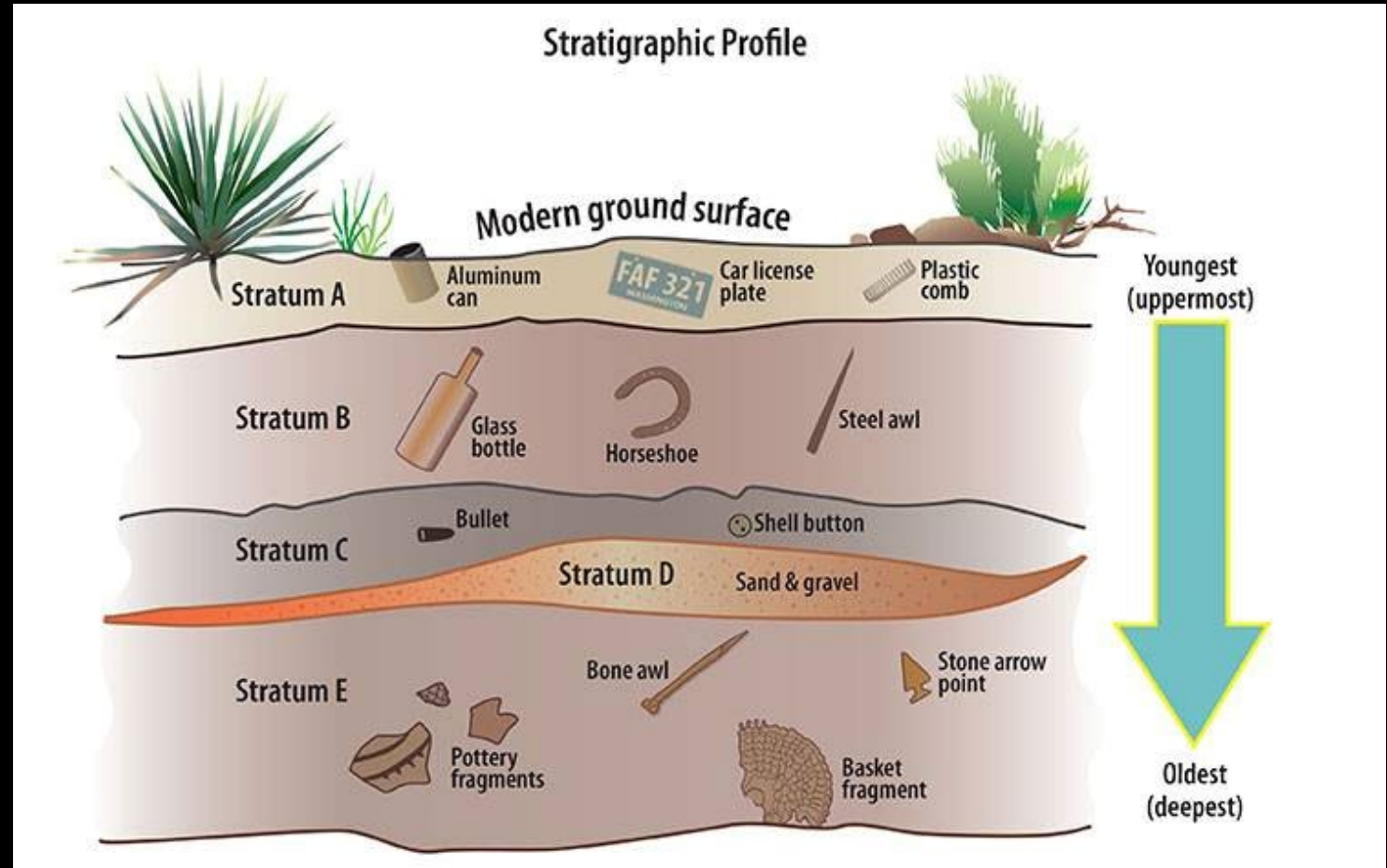
Excavation Methods

- Interpretation:
 - Use stratigraphy and radiocarbon dates to **determine the age of your site** and its occupation
 - **Identify any artifacts** and where they came from: ceramics, beads, tools, etc.
 - **Identify plant and animal remains** to understand what people were eating
 - Pull all this information together to **explain how the ancient people at the site lived**, and what different elements of their culture might have looked like.



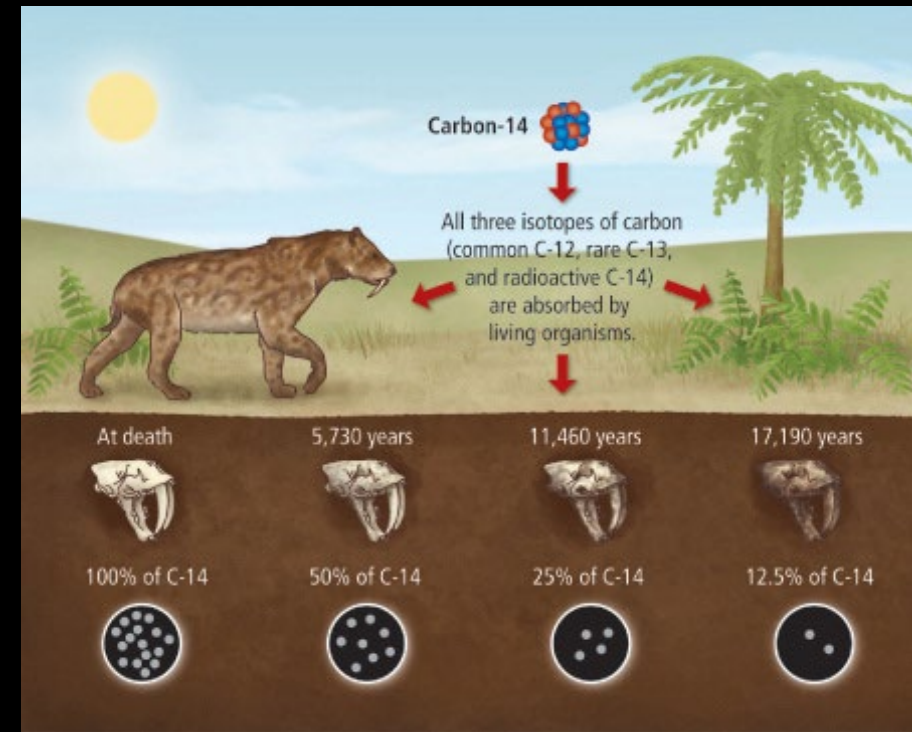
Stratigraphy

- The study of layers of rocks, sediments, and soils, as well as the relative location of features and artifacts.
- The relative position of layers, features, and artifacts, can determine the chronological sequence of their deposition

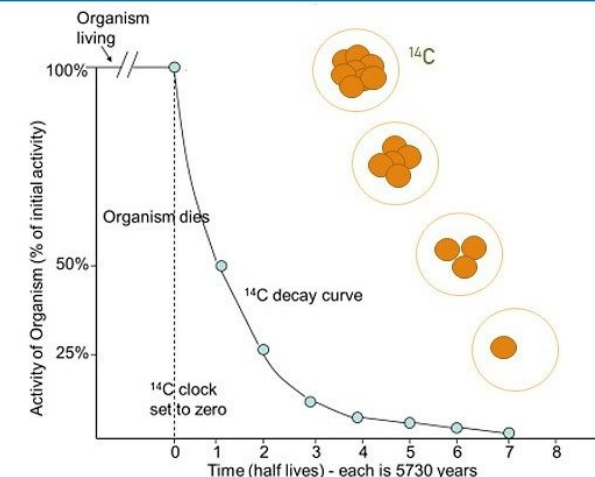


Radiocarbon Dating

- Also called **carbon-14 dating**, this is a method for determining the length of time between the death of an organism and the present.
- This method allows archaeologists to **date archaeological finds** like burned bones, carbonized plants, and human skeletons, among others.



Radiocarbon Formation & Decay



Paleoethnobotany

- Also called archaeobotany
- The field of research that studies plant remains in archaeological contexts
- Includes **macrobotanical remains** (seeds, wood charcoal, plant parts)
- **Microbotanical remains** (starch, phytoliths, and pollen)



Macrobotanical

large

Wood Charcoal

carbonized wood



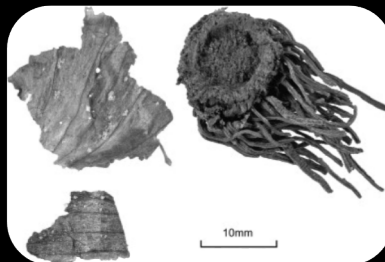
Seeds

wheat seeds



Plant Parts

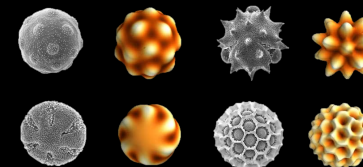
dessicated onion
skins and bulb roots



Microbotanical

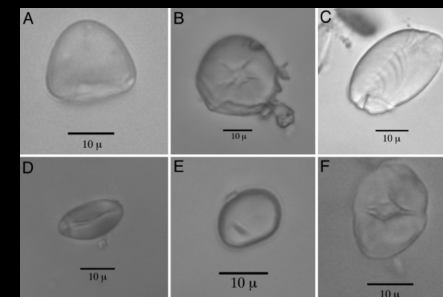
small

Pollen Grains



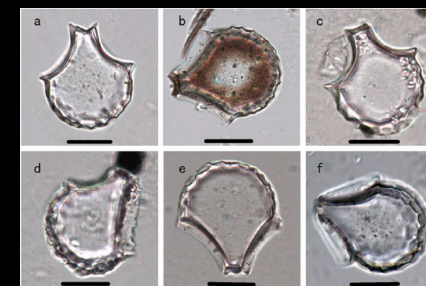
SEM image of pollen

Starch Granules



gourd and squash
starches

Phytoliths



rice phytoliths

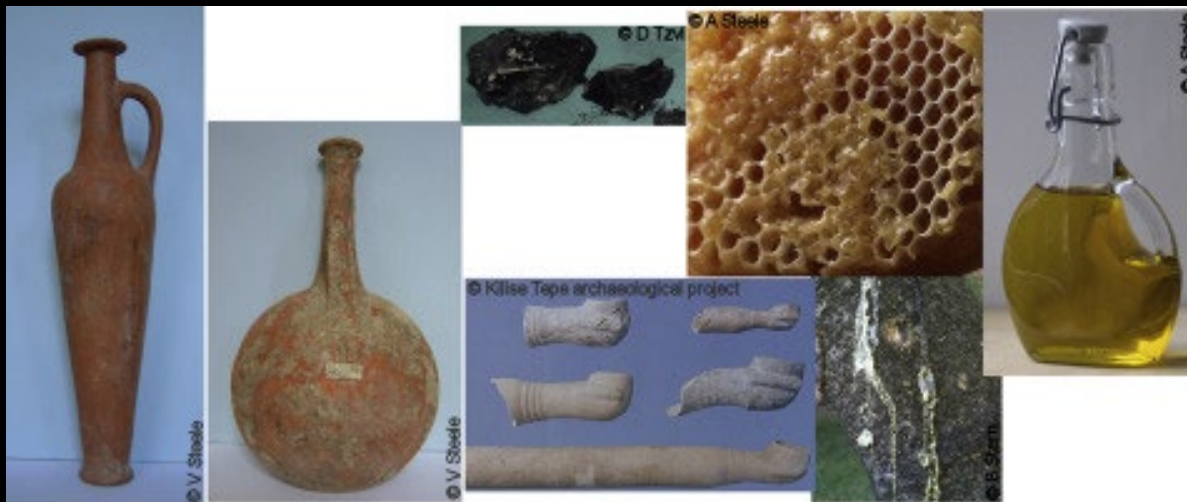
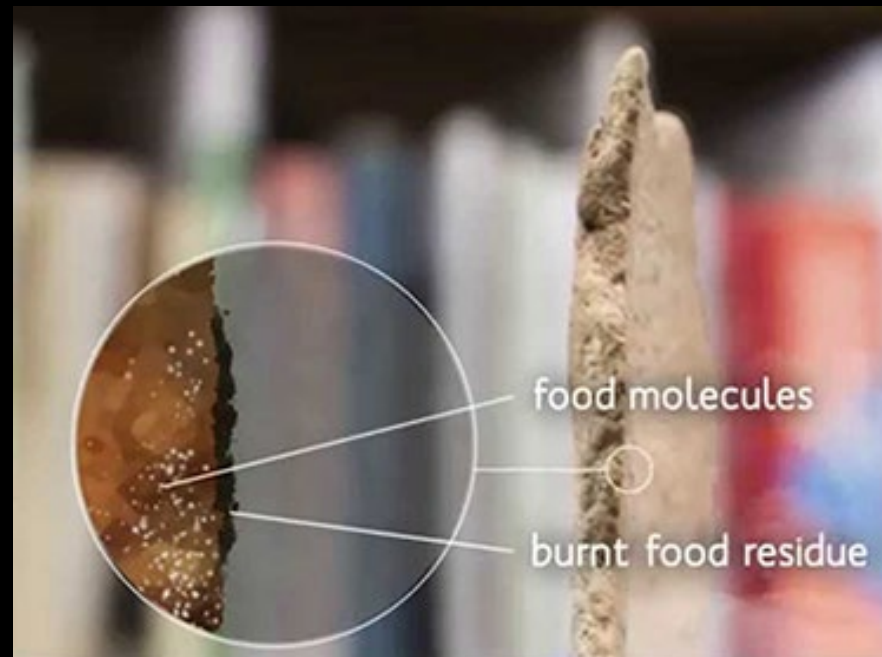
Paleoethnobotany Methods

- Macrobotanical (large remains, don't need a microscope to see)
- Microbotanical (small remains, can't see them without a microscope)
- **Flotation**: Process by which archaeologists use water to separate plant material like carbonized wood and seeds, from soil and heavy artifacts like ceramics and bone
 - Main method by which paleoethnobotanical remains are collected across the world
- **Residue analysis**: Typically used to recover microbotanical remains like starch and phytoliths; involves **washing artifacts like pottery and stone tools** and then extracting and identifying the microbotanical remains



Chemical Residue Analysis

- Field of research that studies the **chemical traces** of human activity in the past
- Most residue analysis is conducted on artifacts, **especially porous pottery** that might absorb the chemicals of the items cooked or stored within the pot



Week 2: Evolutionary History of the Human Diet

Learning Objectives:

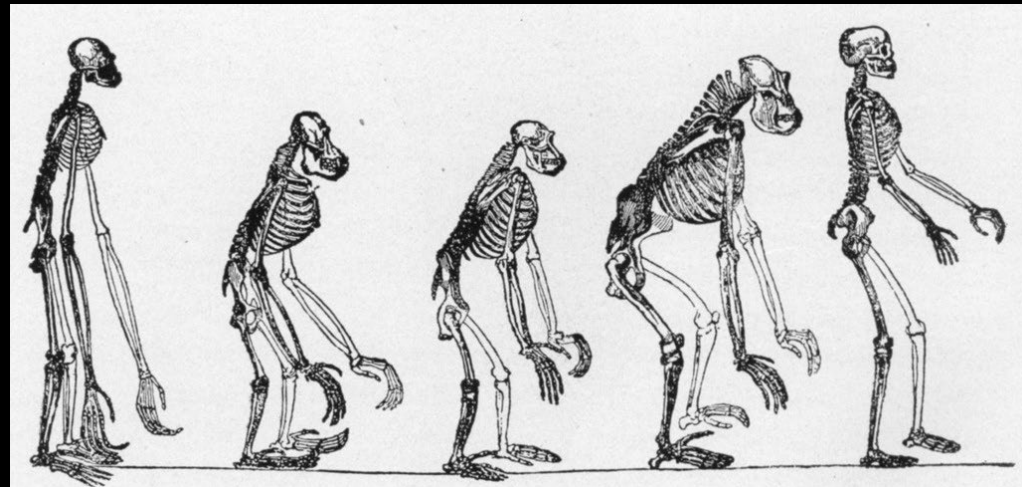
- Understand how the evolutionary history of primate and hominin diet contextualizes dietary options for the modern human species
- Place the human diet of the Pleistocene within its climatic and ecological context
- Assess the evolutionary claims made by proponents of the Paleo Diet
- Describe basic methods for the recovery, identification, and interpretation of animal bones from archaeological sites

Week 2: Key Terms

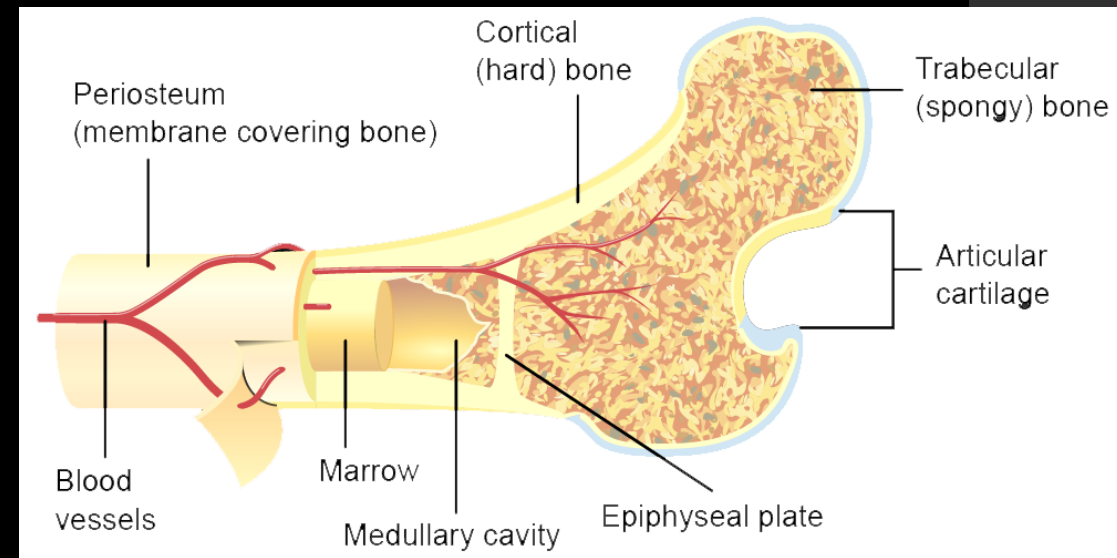
- Bipedal
- Collagen
- Comparative approach
- Cortical bone
- Diaphysis
- Epiphyses
- Hearths
- Heuristic models
- Holocene
- Hominin
- Hybrids
- Hydroxyapatite
- Marrow
- Micromorphology
- MNI
- Monogastric
- Neolithic
- NISP
- Paleolithic
- Pleistocene
- Pleistocene megafauna
- Primary data
- Primates
- Rubefaction
- Secondary data
- Spectroscopy
- Taphonomy
- Taxon
- Trabecular bone
- Zooarchaeology

Comparative Approach

- Used by biological anthropologists and paleoanthropologists to **understand the relationship between anatomical features and behaviors** by drawing on the shared history of evolutionary adaptations to environmental constraints and opportunities between the species under investigation
- Attempts to **explain the similarities and differences among people** within the context of humanity as a whole



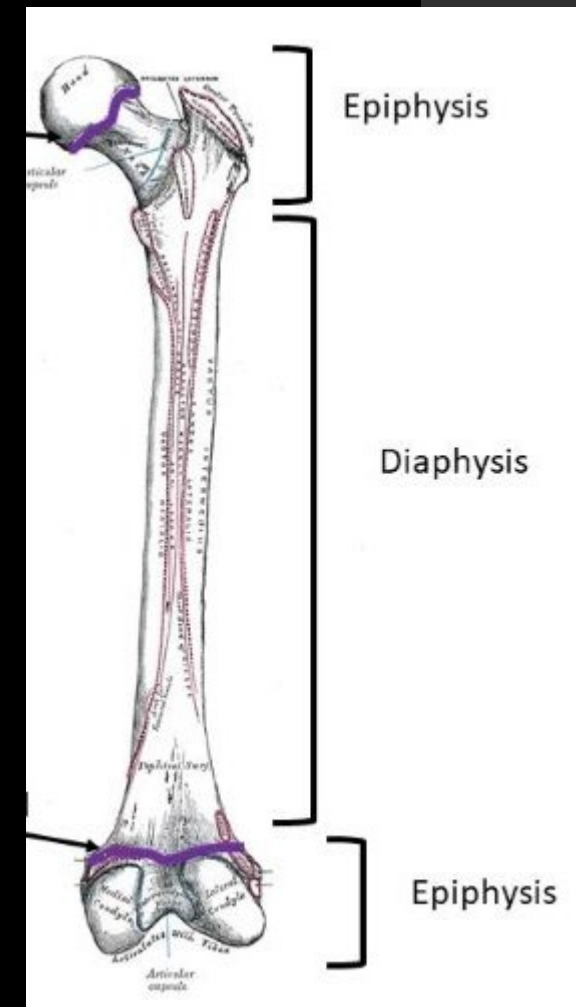
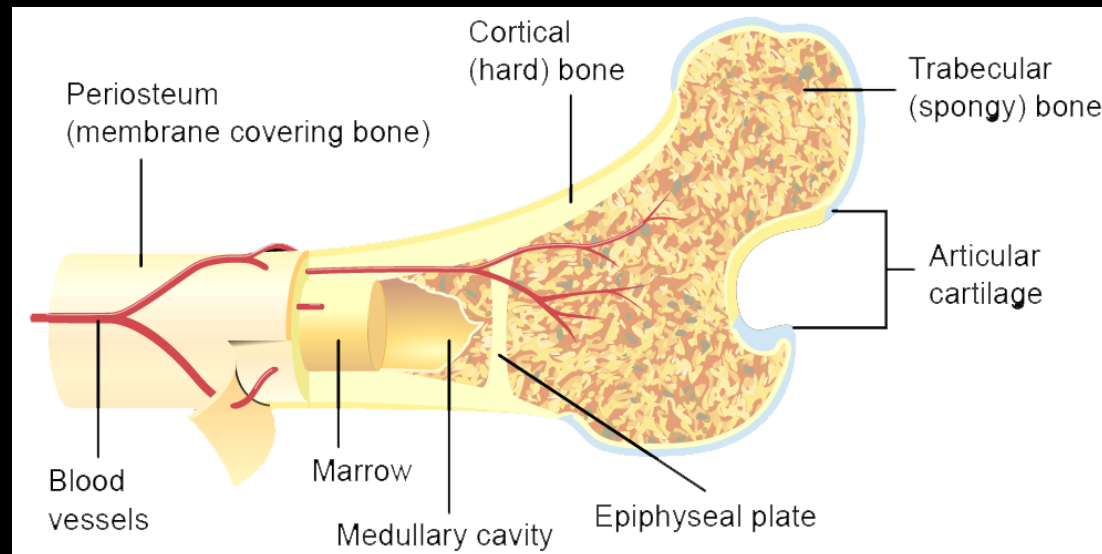
Bone Composition



- **Collagen:** the protein component, also found in cartilage, gives bones their flexibility
- **Cortical bone:** Hard, dense bone found on the outer surface
- **Hydroxyapatite:** The inorganic component, composed of calcium and phosphate, gives bones their strength
- **Marrow:** fatty tissue within bones that produces blood cells.
- **Medullary cavity:** The hollow part of bone that contains bone marrow.
- **Periosteum:** A thin, dense membrane that covers the outer surface of bones

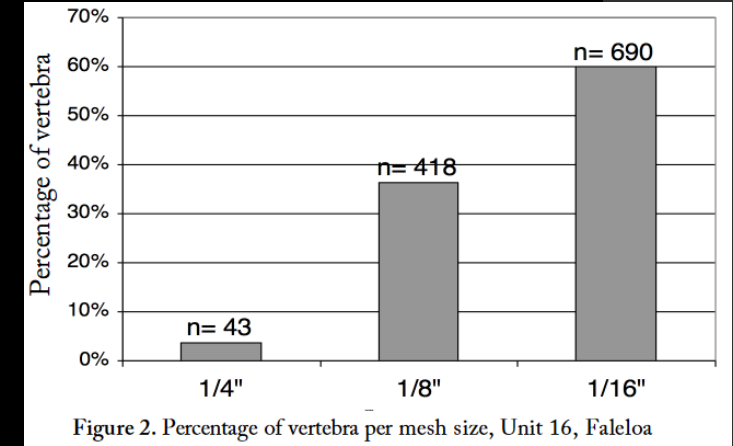
Long Bone Composition

- **Diaphysis:** The shaft of long bones
- **Epiphyses:** The ends of long bones that connect with other bones. Are composed of collagen and fuse with diaphysis later in life, can be used to estimate age
- **Trabecular bone:** Spongy, brittle bone, found in the ends of long bones (epiphyses) or their interior cavities



Zooarchaeology Methods

- **NISP**: “Number of Identified Specimens”, aka count of all bones/bone fragments in a sample/unit/site
- **MNI**: “Minimum Number of Individuals”, the smallest number of animals that could have resulted in the bones in the archaeological sample
- **Screening soil** in the field:
 - Large screen size (1/4 inch)
 - Medium screen size (1/8 inch)
 - Small screen size (1/16 inch)
- Large screens can allow small fish or bird bones to pass through, incorrectly making it look like your sample has less fish/birds than it really does
- Can **bias** what your assemblage looks like!



Micromorphology

- Geological techniques used to study the **microscopic structure** of sediments and soils
- A block of soil is taken to a lab, **hardened with resin**, and extremely thin **slices are taken off**
- These slices are then **studied under the microscope**; possible to identify specific minerals, materials, and depositions



Looking to the Week Ahead!

- Food Journal Essay 2: Sunday, July 12, 11:59 PM ET (Boston time)
- Quiz 1: Due Sunday, July 12, 11:59 PM ET (Boston time)
- Food Journal Presentations due Sunday, July 12, 11:59 PM ET (Boston time)
- Food Journal Presentation Responses/Comments on Week 1 Presentations due Sunday, July 12, 11:59 PM ET (Boston Time)
- Start thinking about your Paper Proposals (due Sunday, July 19)
 - Assignment itself is short, but start thinking about topics that might interest you, specific foods, drinks, or drugs, but **remember that they must be archaeological in focus!**

Food Journal Essay – Learn from Feedback

Before you submit Food Journal Essay #2

- Review the feedback from your facilitator on Food Journal Essay #1
- Use that feedback to avoid repeating the same mistakes

Common things to check:

- Follow required formatting
- Make sure you include all required assignment components (e.g., Works Cited, recipe, appendix, illustrations/Images, etc.)
- When in-text citation and a Works Cited section are required, make sure to use Chicago author-date citation
- Include scientific names of plants/animals at their first mention, for example:
 - maize (*Zea mays*)
 - sheep (*Ovis aries*)
- Proofread for grammar, spelling, and clarity

Questions?